

L12: Degree Dynamics in PA Model

How does the degree of a node change over time in the PA model, as the network grows?

An analytical approach that simplifies the problem considerably is to make two approximations:

a) ignore the discrete-time increments of the model and use a “continuous-time approximation” instead,
and

b) ignore the probabilistic nature of the model and consider a deterministic growth process in which the degree of all nodes increases in a continuous manner based on the PA formula.

Specifically, consider a node i with degree k_i at time t . The rate at which the degree of that node increases at time t is:

$$\frac{dk_i}{dt} = m \Pi(k_i) = m \frac{k_i}{2mt}$$

The differential equation becomes:

$$\frac{dk_i}{k_i} = \frac{dt}{2t}$$

By integrating both sides, we get that:

$$\ln k_i = \frac{1}{2} \ln t + c$$

where c is a constant. Exponentiating both sides:

$$k_i = t^{1/2} e^c$$

The initial condition is that $k_i(t_i) = m$, where t_i is the time instant that node- i is born.

So, the degree of node i increases with time as follows, for $t > t_i$:

$$k_i(t) = m \left(\frac{t}{t_i} \right)^{1/2}$$

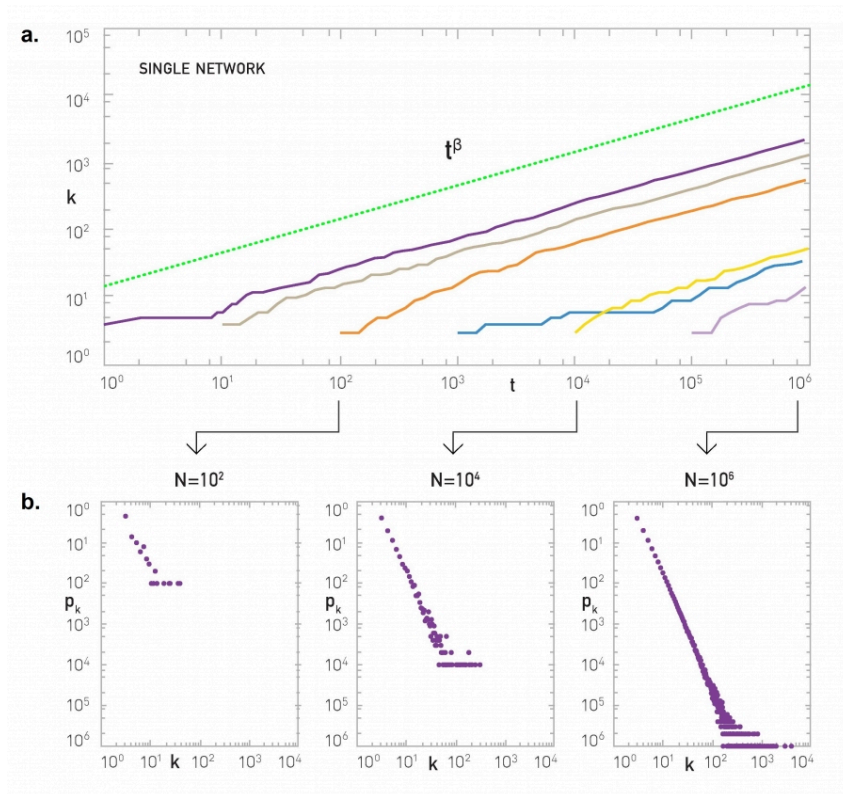
This simple derivation predicts a couple of interesting facts:

First, the degree of all nodes is expected to increase with the square-root of time, i.e., sublinearly. The sublinearity is expected because each new node brings the same number of links m but it has to distribute those links to a growing number of nodes.

Second, older nodes (nodes that were added earlier in the model), accumulate a larger degree over time. In fact, the maximum degree is expected to be the degree of the first node added in the network, i.e., $k_{max}(t) \approx \sqrt{t}$. So, the PA model can capture the “first-mover advantage” that is often seen in economy, especially when companies compete for new products or services.

This last point also highlights a shortcoming of the PA model: it cannot capture that different nodes may have different “attractiveness” for new links. The only node feature that matters is the time at which the node is born.

There are several variations of the PA model that introduce additional node parameters (such as a “quality” factor for each node), and/or different network processes (such as removal of existing nodes or edges, or rewiring of existing edges).



The visualization shows numerical results for nodes born at different times (purple for the node born at $t=1$, orange for the node born at $t=100$, etc). All curves increase approximately with an exponent of $\beta = \frac{1}{2}$ (note the log scale of the x and y axes).

Of course there are statistical fluctuations because these numerical results do not use the deterministic approximation of the previous derivations. The green curve represents the function $t^{1/2}$.

The lower plots show the degree distribution at three different snapshots of the growth process.

Food For Thought

Use the same deterministic and continuous-time approach to derive the degree of a node as a function of time under the following two scenarios:

- Consider the scenario when we have network growth but no preferential attachment. Suppose that the network grows by one node at a time but the new node adds m links to randomly chosen nodes, instead of using preferential attachment.
- Consider the scenario when we have preferential attachment but no network growth. Suppose that the network size remains constant (equal to N nodes). At each time step, a node is selected randomly and connects to a node i of degree k_i with the PA probability $\Pi(k_i)$. If a node does not have any edges, we set arbitrarily that $k=1$ so that it can potentially get some edges.